

Operating Loci — Generator Load k Sets the Equilibrium

V10-Spoke · 11 m/s · standard frame · one colour per blade family · point label = $k_{\text{mppt}} \cdot P_{\text{gen}} = k \cdot \omega^3$

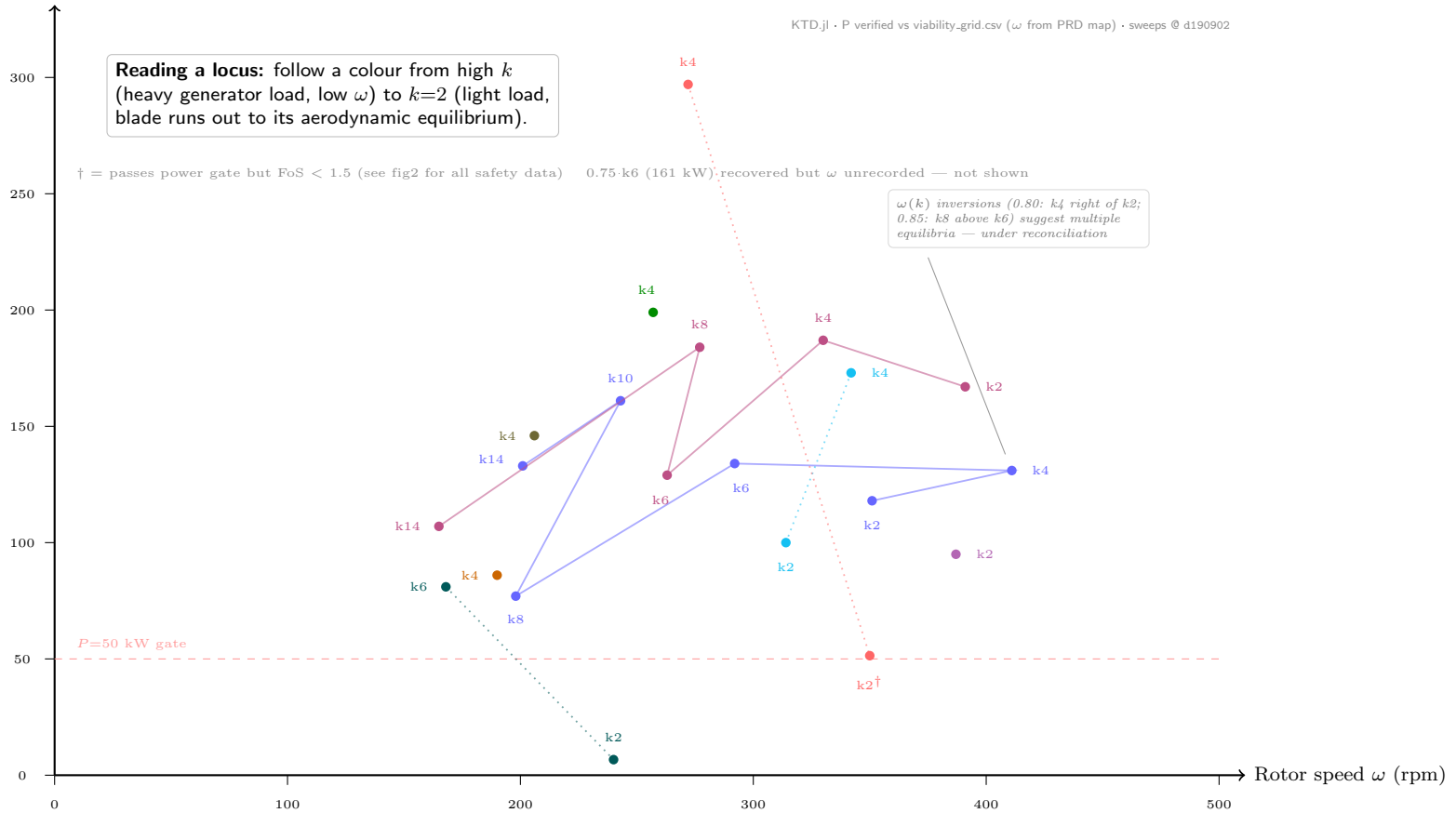
Ground power P (kW)

KTD.jl · P verified vs viability_grid.csv (ω from PRD map) · sweeps @ d190902

Reading a locus: follow a colour from high k (heavy generator load, low ω) to $k=2$ (light load, blade runs out to its aerodynamic equilibrium).

† = passes power gate but FoS < 1.5 (see fig2 for all safety data) 0.75 k6 (161 kW) recovered but ω unrecorded — not shown

$\omega(k)$ inversions (0.80: k4 right of k2; 0.85: k8 above k6) suggest multiple equilibria — under reconciliation



Blade family λ :

0.69 0.75 0.80 0.85 0.90 0.95 1.00 1.05 1.10

Solid line = full k sweep (kickstart-recovered families) Dotted = 2-point family Singles = only one standard-frame operating point known